

Lyapunov-Controlled Minimax Policy Learning for Adversarial Dynamic Spectrum Access

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Abstract—Adaptive jamming makes dynamic spectrum access (DSA) a sequential zero-sum Markov game: both channel selections change packet service and hence the future queue state. We propose Lyapunov-Controlled Minimax Policy Learning (LCMPL) for this queue-aware game. LCMPL combines the simultaneous policy-gradient field with a Jacobian-field correction, gives the two directions separate step sizes, and regulates them using local game geometry and a communication-aware Lyapunov merit. A curvature identity characterizes when the correction dissipates rotational saddle-field energy. Under local smoothness, geometric coverage, and bounded learning and local-model errors, the time-averaged expected stationarity residual has an $O(1/K)$ transient plus an explicit residual term over K joint updates. A local policy-gap condition further converts this result into a bound on the transmitter’s worst-case queue-aware utility loss. Experiments show statistically resolved improvements over the field-only ablation in worst-case utility, exploitability, goodput, and backlog; load sweeps further show that these gains grow under heavier traffic.

Index Terms—Anti-jamming communications, dynamic spectrum access, reinforcement learning, minimax policy learning, zero-sum Markov games.

I. INTRODUCTION

In dynamic spectrum access (DSA), a secondary transmitter selects a channel under incumbent activity and time-varying propagation [1]; practical access decisions must also account for switching delay, traffic backlog, interference, and jamming [2], [3], [4], [5]. Adaptive jamming couples radio variation with opponent adaptation. More importantly, a poor joint channel choice can prevent packet service, enlarge the queue, and change the next decision state. The learning problem is therefore not a sequence of disconnected static games: its optimization dynamics directly affect future communication congestion.

A two-player zero-sum Markov game captures this interaction [6], [7]. The transmitter policy π_θ maximizes a discounted utility that rewards delivered packets and penalizes backlog, packet loss, and channel switching, while the jammer policy ν_ψ minimizes it. Because their channel actions determine the service distribution, both players affect the queue transition as well as the immediate utility. Their simultaneous policy gradients consequently form a coupled saddle field rather than a single improvement direction.

The main learning difficulty is rotational coupling. As Fig. 1 shows, gradient descent-ascent (GDA) can cycle in a bilinear local model, whereas the Jacobian-field action supplies radial contraction [8]. The useful balance changes with local coupling,

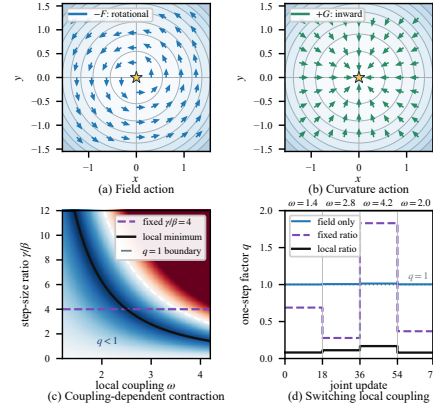


Fig. 1. Why saddle-field step sizes must adapt to local coupling. For $\mathbf{z}_{k+1} = \mathbf{z}_k - \beta \mathbf{F}_\omega + \gamma \nabla \mathbf{F}_\omega \mathbf{F}_\omega$, ω is coupling strength and $q_k = \|\mathbf{z}_{k+1}\|/\|\mathbf{z}_k\|$ is the one-step norm ratio. (a)–(b) show rotation and correction; (c) shows that $q_k < 1$ and the minimizing γ/β vary with ω ; (d) shows a fixed ratio becoming expansive while the local ratio remains contractive.

so a fixed curvature-to-field step-size ratio can move from contraction to expansion as queue occupancy, radio conditions, and policies evolve.

The proximal point method (PPM) and extragradient method (EGM) suppress rotation by evaluating an implicit or extrapolated field [9], [10]. Their local expansions share a Jacobian-field correction but couple its effective scale to the field step through a single step size [11]. Adaptive Mirror-Prox methods tune extragradient updates from observed operator geometry [12], and double-step extragradient separates the extrapolation and update stages [13]. These methods establish the value of adaptive scaling; the remaining design question is how to balance the field and curvature coordinates as radio conditions and policies change. Field-energy reduction alone can also favor a low-utility stationary policy, so it is not a sufficient criterion by itself.

In this paper, we propose Lyapunov-Controlled Minimax Policy Learning (LCMPL). Following Lyapunov-drift step-size control [14], LCMPL assigns separate step sizes to the field and Jacobian-field directions and selects them through a two-variable quadratic program (QP) that predicts local merit decrease. The merit connects policy stationarity with worst-case queue-aware communication utility, while a same-state utility test rejects detrimental curvature. We establish an $O(1/K)$ stationarity transient with an explicit learning-error residual and, under a local policy-gap condition, a worst-case utility bound. Queue-aware experiments compare LCMPL with

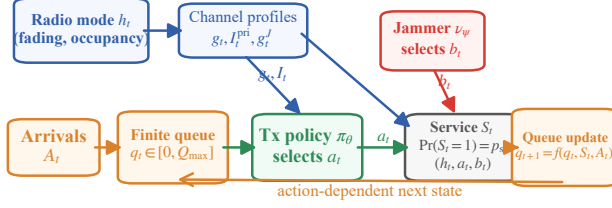


Fig. 2. Queue-aware adversarial DSA: both channel actions change packet service and the next queue state.

representative simultaneous-gradient, extragradient, implicit-update, and clipped-policy baselines and examine how its field-curvature gain changes from light to heavy traffic.

II. ADVERSARIAL DSA SYSTEM AND MARKOV-GAME MODEL

A. Queue-Aware Wireless Interaction

Fig. 2 considers a secondary transmitter with a finite buffer, its receiver, incumbent activity, and an adaptive jammer over M subchannels. At slot t , an exogenous mode $h_t \in \mathcal{H} = \{1, \dots, L\}$ selects a correlated radio profile $x_t = (g_t, I_t^{\text{pri}}, g_t^J)$: secondary-link gains, incumbent interference, and jammer-receiver gains across channels. The transmitter and jammer simultaneously choose $a_t \in \mathcal{A}$ and $b_t \in \mathcal{B}$, where $\mathcal{A} = \mathcal{B} = \{0, \dots, M-1\}$. The state

$$s_t = (h_t, x_t, q_t, a_{t-1}) \in \mathcal{S}$$

also contains queue length $q_t \in \{0, \dots, Q_{\max}\}$ and the previous transmitter channel a_{t-1} . Thus h_t describes the external radio regime, while q_t records the endogenous consequence of earlier joint actions.

Let P_s , P_J , and N_0 denote transmission power, jamming power, and receiver noise, respectively. The factor $\ell(a, b)$ describes co-channel and adjacent-channel leakage, and $c_{\text{sw}} \geq 0$ is the channel-switching cost. For actions (a_t, b_t) , the mean signal-to-interference-plus-noise ratio (SINR) is

$$\bar{\Gamma}_t(a_t, b_t) = \frac{P_s g_t(a_t)}{N_0 + I_t^{\text{pri}}(a_t) + P_J g_t^J(b_t) \ell(a_t, b_t)}. \quad (1)$$

One head-of-line packet carries R_0 bit/s/Hz. Under block Rayleigh fading, its service indicator $S_t \in \{0, 1\}$ obeys

$$\Pr(S_t = 1 \mid s_t, a_t, b_t) = \mathbf{1}_{\{q_t > 0\}} \exp \left[-\frac{2^{R_0} - 1}{\bar{\Gamma}_t(a_t, b_t)} \right].$$

Independent Bernoulli arrivals $A_t \sim \text{Bernoulli}(\lambda_A)$ enter at the end of the slot. The dropped-packet count and finite-buffer update are

$$\begin{aligned} D_t &= [q_t - S_t + A_t - Q_{\max}]^+, \\ q_{t+1} &= \min\{Q_{\max}, [q_t - S_t]^+ + A_t\}. \end{aligned} \quad (2)$$

The stage utility rewards delivered goodput and penalizes congestion, dropping, and switching:

$$r_t = R_0 S_t - \alpha_q q_t - \alpha_d D_t - c_{\text{sw}} \mathbf{1}_{\{a_t \neq a_{t-1}\}}, \quad (3)$$

where $\alpha_q, \alpha_d \geq 0$ and $\mathbf{1}_{\{\cdot\}}$ is the indicator function.

B. Radio Dynamics and Minimax Policy Objective

The exogenous mode follows $P_h(h'|h)$, but the complete state transition is action dependent. Let $P_Q(q'|q, h, a, b)$ be the queue kernel induced by the service and arrival laws above. Writing $s' = (h', x', q', a'_{\text{prev}})$, the transition factors as

$$P(s' \mid s, a, b) = P_h(h'|h) P_Q(q'|q, h, a, b) \mathbf{1}_{\{a'_{\text{prev}}=a\}},$$

with x' determined by h' . Because P_Q depends on $\bar{\Gamma}_t(a, b)$, both players alter future congestion through their channel choices; the model is therefore a genuine stochastic Markov game rather than a repeated static game. We use $P_h(h|h) = \rho_h$ and $P_h(h'|h) = (1 - \rho_h)/(L - 1)$ for $h' \neq h$.

Let $\pi_\theta(a|s)$ and $\nu_\psi(b|s)$ denote differentiable transmitter and jammer policies. The physical normalized discounted queue-aware utility is

$$J(\theta, \psi) = (1 - \delta) \mathbb{E} \left[\sum_{t=0}^{\infty} \delta^t r_t \right],$$

where $\delta \in (0, 1)$ is the discount factor. We train with the entropy-regularized return

$$J_\tau = J + (1 - \delta) \tau \mathbb{E} \left[\sum_{t \geq 0} \delta^t \{H(\pi_\theta(\cdot|s_t)) - H(\nu_\psi(\cdot|s_t))\} \right],$$

where $\tau > 0$ is the entropy weight and $H(\cdot)$ is the Shannon entropy. Training directions and the Lyapunov model use J_τ ; acceptance and reported communication metrics use the physical utility J . Let Π_T and Π_J be the full stationary Markov policy spaces. The transmitter's worst-case utility against a stationary best response (BR) is

$$U_{\text{BR}}(\theta) = \min_{\nu \in \Pi_J} J(\pi_\theta, \nu).$$

The unregularized exploitability is

$$\Delta(\mathbf{z}) = \max_{\pi \in \Pi_T} J(\pi, \nu_\psi) - \min_{\nu \in \Pi_J} J(\pi_\theta, \nu).$$

Replacing J by J_τ defines the regularized gap Δ_τ used inside the merit. Dynamic programming computes U_{BR} , Δ , and Δ_τ . We additionally report goodput, average backlog, and drop probability against the utility-minimizing BR.

III. GEOMETRY-AWARE MINIMAX POLICY LEARNING

A. Curvature-Induced Bi-Direction Learning

For the joint parameter $\mathbf{z} = (\theta, \psi)$, define the saddle field

$$\mathbf{F}(\mathbf{z}) = \begin{bmatrix} -\nabla_\theta J_\tau(\theta, \psi) \\ \nabla_\psi J_\tau(\theta, \psi) \end{bmatrix}. \quad (4)$$

For a scalar step size $s > 0$, let $\mathbf{z}_{k+1}$ denote the next joint policy parameter. The field energy $\phi(\mathbf{z}) = \|\mathbf{F}(\mathbf{z})\|^2/2$ measures joint policy stationarity. Under GDA, $\phi(\mathbf{z}_{k+1}^{\text{GDA}}) = \phi(\mathbf{z}_k) - s \langle \nabla \phi(\mathbf{z}_k), \mathbf{F}(\mathbf{z}_k) \rangle + O(s^2)$; pure rotation has zero first-order change despite $\mathbf{F}(\mathbf{z}_k) \neq 0$. The representative GDA [15], PPM [9], and EGM [10] updates are

$$\begin{aligned} \text{GDA: } \mathbf{z}_{k+1}^{\text{GDA}} &= \mathbf{z}_k - s \mathbf{F}(\mathbf{z}_k), \\ \text{PPM: } \mathbf{z}_{k+1}^{\text{PPM}} &= \mathbf{z}_k - s \mathbf{F}(\mathbf{z}_{k+1}^{\text{PPM}}), \\ \text{EGM: } \tilde{\mathbf{z}}_k &= \mathbf{z}_k - s \mathbf{F}(\mathbf{z}_k), \quad \mathbf{z}_{k+1}^{\text{EGM}} = \mathbf{z}_k - s \mathbf{F}(\tilde{\mathbf{z}}_k). \end{aligned}$$

Here, $\tilde{z}_k$ is EGM's extrapolation point; GDA is explicit, PPM uses an implicit next-policy field, and EGM uses an explicit predictor. Let $\nabla F(z)$ denote the Jacobian of the saddle field. Expanding the displaced field in EGM, or the implicit field in PPM, gives $z_{k+1} = z_k - sF(z_k) + s^2\nabla F(z_k)F(z_k) + O(s^3)$. Their common correction motivates the Jacobian-field direction

$$G(z) = \nabla F(z)F(z). \quad (5)$$

The term $+s^2G$ is the shared local correction that distinguishes PPM and EGM from the rotational field step. It points inward in the bilinear model of Fig. 1 and motivates treating $-F$ and $+G$ as separate update coordinates. PPM and EGM couple their effective scales through the fixed relation (s, s^2) , although the useful balance changes with local coupling. LCMPL instead assigns the field and curvature coordinates separate step sizes β_k and γ_k :

$$z_{k+1} = z_k - \beta_k F(z_k) + \gamma_k G(z_k). \quad (6)$$

Setting $\gamma_k = 0$ recovers GDA, whereas $\beta_k = s$ and $\gamma_k = s^2$ match the leading PPM and EGM correction. For later use, write $F_k = F(z_k)$ and $G_k = G(z_k)$. The finite-game experiments evaluate these directions exactly; in sampled implementations, automatic differentiation forms G_k as a Jacobian-vector product. The curvature direction introduces no additional wireless action; it modifies only the policy update using local game geometry. The remaining design problem is to choose β_k and γ_k without trading policy stabilization for communication performance.

B. Communication-Aware Lyapunov Policy Update

A small field energy identifies joint policy stationarity, while communication robustness additionally depends on the attained worst-case utility. LCMPL therefore evaluates both directions through a Lyapunov merit that combines field energy, exploitability, and worst-case-utility deficit. Define the normalized communication loss

$$\mathcal{L}_c(z) = \lambda_\Delta \left(\frac{\Delta_\tau(z)}{\Delta_{\tau,0}} \right)^2 + \lambda_U \left(\frac{D_{U,\tau}(\theta)}{D_0} \right)^2,$$

where $\lambda_\Delta, \lambda_U \geq 0$ are merit weights and $\Delta_{\tau,0}, D_0 > 0$ are initial scales. The first term penalizes a policy pair that remains exploitable, while the second penalizes loss of robust communication utility. Let $r_{\max}$ be the largest stage utility and define $U_{\text{ub},\tau} = r_{\max} + \tau \log M$. The deficit $D_{U,\tau} = U_{\text{ub},\tau} - \min_{\nu \in \Pi_J} J_\tau(\pi_\theta, \nu) \geq 0$ uses this ceiling for normalization. The Lyapunov merit is

$$V(z) = \frac{\phi(z)}{\phi_0} + \mathcal{L}_c(z), \quad (7)$$

where $\phi_0 > 0$ is the initial field-energy scale. The merit controls training, and the physical utility remains the communication objective used for acceptance and reporting. Its one-step drift provides a common criterion for geometric stabilization and communication quality.

Let $\mathcal{F}_k$ contain the learning history before update k . The update lies in the bi-direction subspace $d_{k,1} = -F_k$, $d_{k,2} =$

G_k ; sampled-direction discrepancies enter the model-error term below. For a candidate $u = (\beta, \gamma)^\top$, define its conditional Lyapunov drift by

$$\mathcal{D}_k(u) = \mathbb{E}[V(z_k + \beta d_{k,1} + \gamma d_{k,2}) - V(z_k) \mid \mathcal{F}_k].$$

A negative $\mathcal{D}_k(u)$ means that the candidate step decreases the expected merit given the information available before update k . Minimizing this drift therefore balances three goals in one decision: reducing the saddle-field energy, limiting exploitability, and preserving the transmitter's worst-case utility. Because $V \geq 0$, such one-step decreases can also be summed over time, which is the basis of the finite-time bound in Section IV. Directly minimizing $\mathcal{D}_k$ would require repeated evaluation of the nonlinear policy objective for every step-size pair. Instead, a directional Taylor expansion gives

$$\begin{aligned} \mathcal{D}_k(u) &= \sum_{i=1}^2 u_i \mathbb{E}[\langle \nabla V(z_k), d_{k,i} \rangle \mid \mathcal{F}_k] \\ &\quad + \frac{1}{2} \sum_{i,j=1}^2 u_i u_j \mathbb{E}[d_{k,i}^\top \nabla^2 V(z_k) d_{k,j} \mid \mathcal{F}_k] + \rho_k(u). \end{aligned}$$

where ρ_k collects higher-order and direction-estimation errors. LCMPL approximates the field- and curvature-direction slopes and curvatures in this expansion and absorbs the resulting discrepancy, including ρ_k , into the model-error term used in Section IV. This yields the local surrogate

$$\begin{aligned} m_k(u) &= a_k^\top u + \frac{1}{2} u^\top H_k u, \\ \tilde{u}_k &= \arg \min_{u \in \mathcal{U}_k} m_k(u), \quad \mathcal{U}_k = [0, \bar{\beta}] \times [0, \bar{\gamma}], \end{aligned} \quad (8)$$

where $a_k \in \mathbb{R}^2$ contains directional slopes, $H_k \succeq h_{\min} I$ is a regularized diagonal directional-curvature estimate, I is the 2×2 identity matrix, $h_{\min} > 0$, and $\bar{\beta}, \bar{\gamma} > 0$ cap the field and curvature step sizes. The box keeps the Taylor model local, while the positive curvature regularization makes the two-variable problem well posed. The QP therefore replaces an expensive search over the exact drift by a tractable prediction of the best local merit decrease. When rotational coupling makes the field coordinate ineffective, a negative curvature slope can activate γ_k and move the update inward. When the field coordinate already decreases the merit, the same model can reduce or suppress the curvature step. LCMPL compares the proposed pair with the field-only minimizer of the same model and accepts curvature only when the realized merit and physical worst-case utility are no worse. LCMPL-F fixes $\gamma_k = 0$ while retaining the same field-direction merit, local model, and acceptance rule. Fig. 3 summarizes how queue-aware feedback determines the bi-direction update and how the Lyapunov-drift model selects its field and curvature step sizes.

IV. LEARNING STABILITY AND COMMUNICATION GUARANTEES

This section connects curvature-induced field-energy decrease to the implemented update and its finite-time communication guarantee. Write $\nabla F = S + W$, where $S = (\nabla F + \nabla F^\top)/2$ and $W = (\nabla F - \nabla F^\top)/2$.

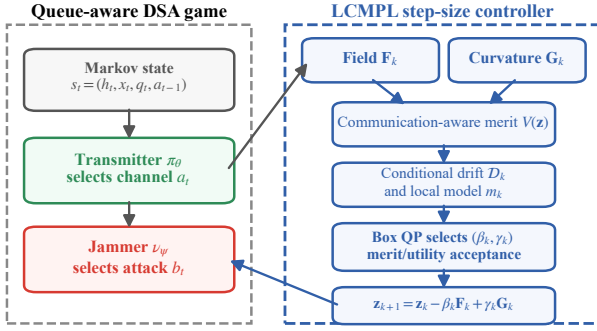


Fig. 3. LCMPL minimax policy-learning framework.

Proposition 1 (Curvature identity for adversarial policy learning): At every point where $\mathbf{F}$ is differentiable, the Jacobian–field direction satisfies

$$\langle \nabla \phi, \mathbf{G} \rangle = \|\mathbf{S}\mathbf{F}\|^2 - \|\mathbf{W}\mathbf{F}\|^2. \quad (9)$$

Hence, $+\mathbf{G}$ decreases the field energy when $\|\mathbf{W}\mathbf{F}\| > \|\mathbf{S}\mathbf{F}\|$.

The proof is given in Appendix A-A.

Proposition 1 explains why the curvature coordinate can reduce field energy. In particular, antisymmetric coupling creates the rotation of the field step and the inward action of the Jacobian–field correction. The next conditions connect that ideal geometry to the local model and accepted learning update. Define $D_F(\mathbf{z}) = \langle \nabla V(\mathbf{z}), \mathbf{F}(\mathbf{z}) \rangle$ and $D_G(\mathbf{z}) = -\langle \nabla V(\mathbf{z}), \mathbf{G}(\mathbf{z}) \rangle$, the first-order decreases of V along $-\mathbf{F}$ and $+\mathbf{G}$. Write $\Delta V_k = V(\mathbf{z}_{k+1}) - V(\mathbf{z}_k)$.

Assumption 1 (Local regularity and model accuracy):

- (a) A compact set $\mathcal{T}$ contains the accepted iterates and candidate segments, and $\mathbf{F} \in C^1$ and $V \in C^2$ on a neighborhood of $\mathcal{T}$.
- (b) For a fixed $s > 0$, both pure-coordinate pairs se_1 and se_2 belong to $\mathcal{U}_k$, where e_i is the i th coordinate vector.
- (c) Finite nonnegative $\mathcal{F}_k$ -measurable errors ϵ_k^{acc} and ϵ_k^{mdl} satisfy

$$\mathbb{E}[\Delta V_k \mid \mathcal{F}_k] \leq \mathbb{E}[m_k(\tilde{\mathbf{u}}_k) \mid \mathcal{F}_k] + \epsilon_k^{\text{acc}},$$

$$\mathbb{E}[m_k(se_i) \mid \mathcal{F}_k] \leq -sD_i(\mathbf{z}_k) + \epsilon_k^{\text{mdl}}, \quad i \in \{1, 2\},$$

where $D_1 = D_F$ and $D_2 = D_G$.

These errors aggregate policy-gradient and Jacobian-action noise, the directional-model approximation, and the acceptance step. For the exact finite-game updates reported here, they record local-model and acceptance mismatch; sampled implementations additionally contribute estimation error.

Define the field-energy decrease signals

$$\chi_F(\mathbf{z}) = \mathbf{F}^\top \mathbf{S}\mathbf{F}, \quad \chi_G(\mathbf{z}) = -\langle \nabla \phi(\mathbf{z}), \mathbf{G}(\mathbf{z}) \rangle.$$

Positive χ_F and χ_G certify decrease of ϕ along $-\mathbf{F}$ and $+\mathbf{G}$, respectively.

Assumption 2 (Geometric coverage): The local coverage modulus

$$m_{\text{geo}} = \inf_{\mathbf{z} \in \mathcal{T}} \frac{\max\{\chi_F(\mathbf{z}), \chi_G(\mathbf{z})\}}{\|\mathbf{F}(\mathbf{z})\|^2} \quad (10)$$

satisfies $m_{\text{geo}} > 0$.

Geometric coverage holds when either the field or curvature coordinate decreases field energy at a rate proportional to $\|\mathbf{F}\|^2$, allowing rotational points where the field coordinate is not a strict descent direction. Let $\sigma_{\min}$ denote the smallest singular value. At a target stationary point $\mathbf{z}^*$, the conditions $\mathbf{S}(\mathbf{z}^*) \succeq 0$ and $\sigma_{\min}(\nabla \mathbf{F}(\mathbf{z}^*)) > 0$ imply Eq. (10) on a sufficiently small neighborhood by compactness of the unit sphere and continuity; this includes nonsingular pure rotation, where $\chi_F = 0$ and $\chi_G > 0$. Compactness and smoothness make

$$B_c = \sup_{\mathbf{z} \in \mathcal{T}} \max\{|\langle \nabla \mathcal{L}_c, \mathbf{F} \rangle|, |\langle \nabla \mathcal{L}_c, \mathbf{G} \rangle|\} < \infty.$$

Because $V = \phi/\phi_0 + \mathcal{L}_c$, the larger of the field and curvature directional signals satisfies

$$\max\{D_F(\mathbf{z}), D_G(\mathbf{z})\} \geq \mu_{\text{geo}} \|\mathbf{F}(\mathbf{z})\|^2 - B_c, \quad (11)$$

$$\mu_{\text{geo}} = m_{\text{geo}}/\phi_0.$$

Define the total conditional learning residual

$$\mathcal{E}_k(s) = sB_c + \epsilon_k^{\text{acc}} + \epsilon_k^{\text{mdl}}. \quad (12)$$

This single term separates the geometric benefit of the bi-direction update from stochastic sampling, local approximation, and acceptance errors.

Proposition 2 (Conditional Lyapunov drift): Under Assumptions 1–2, the accepted update satisfies

$$\begin{aligned} \mathbb{E}[V(\mathbf{z}_{k+1}) - V(\mathbf{z}_k) \mid \mathcal{F}_k] \\ \leq -s\mu_{\text{geo}} \|\mathbf{F}(\mathbf{z}_k)\|^2 + \mathcal{E}_k(s). \end{aligned}$$

The proof is given in Appendix A-B.

Theorem 1 (Main finite-time minimax stationarity result): Under Assumptions 1–2, every integer $K \geq 1$ satisfies

$$\frac{1}{K} \sum_{k=0}^{K-1} \mathbb{E} \|\mathbf{F}(\mathbf{z}_k)\|^2 \leq \frac{V(\mathbf{z}_0)}{s\mu_{\text{geo}}K} + \frac{\bar{\mathcal{E}}_K(s)}{s\mu_{\text{geo}}}, \quad (13)$$

where $\bar{\mathcal{E}}_K(s) = K^{-1} \sum_{k=0}^{K-1} \mathbb{E} \mathcal{E}_k(s)$.

The proof is given in Appendix A-C.

If $\sup_k \mathbb{E} \mathcal{E}_k(s) \leq \bar{E} < \infty$, Theorem 1 gives an $O(1/K)$ transient plus the floor $\bar{E}/(s\mu_{\text{geo}})$. It covers pure rotation: the field has zero first-order effect, while curvature supplies the strict decrease in Fig. 1 and Proposition 1. Here, s serves as an analytical comparator, whereas β_k and γ_k are the implemented step sizes.

The following local compatibility condition bridges Theorem 1 from neural parameter-space stationarity to full-policy best responses.

Let $v^* = \max_{\pi \in \Pi_T} \min_{\nu \in \Pi_J} J(\pi, \nu)$ denote the unregularized stationary-policy game value. Since each M -ary policy has entropy at most $\log M$, the regularized and unregularized returns differ by at most $\tau \log M$ for either player. Consequently,

$$0 \leq v^* - U_{\text{BR}}(\boldsymbol{\theta}) \leq \Delta_\tau(\mathbf{z}) + 2\tau \log M. \quad (14)$$

Let B_K denote the right-hand side of Eq. (13).

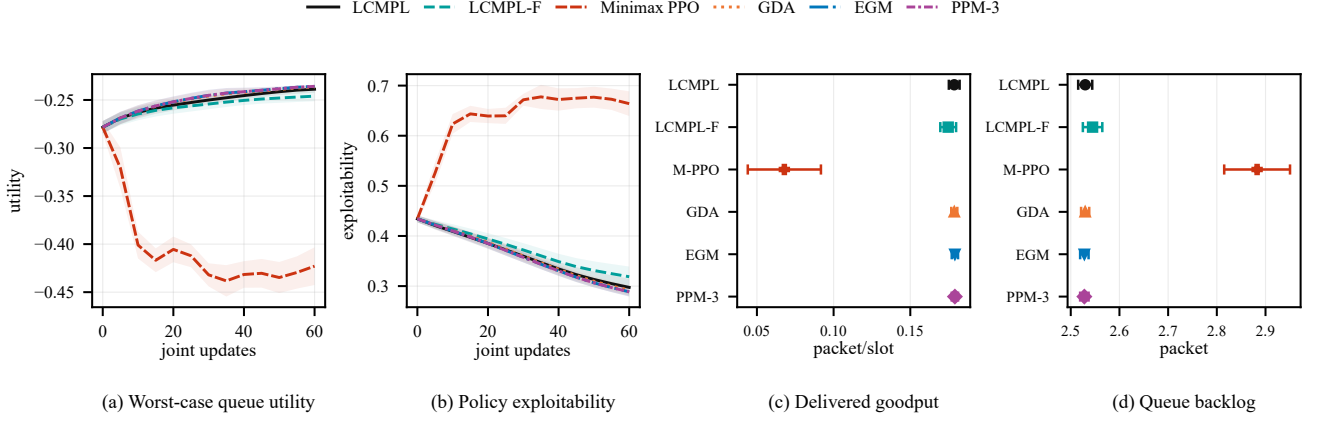


Fig. 4. Queue-aware nominal benchmark at moderate load over 12 seeds. Curves show means with one standard error; points show final means with 95% confidence intervals.

Corollary 1 (Main communication result: worst-case utility):

If nonnegative constants C_{gap} and ξ_{gap} satisfy $\Delta_{\tau}(z) \leq C_{\text{gap}} \|\mathbf{F}(z)\|^2 + \xi_{\text{gap}}$ on $\mathcal{T}$, then

$$\frac{1}{K} \sum_{k=0}^{K-1} \mathbb{E}[v^* - U_{\text{BR}}(\theta_k)] \leq C_{\text{gap}} B_K + \xi_{\text{gap}} + 2\tau \log M.$$

The proof is given in Appendix A-D.

The last term is the entropy-regularization error.

V. COMMUNICATION PERFORMANCE EVALUATION

We evaluate worst-case discounted utility, exploitability, delivered goodput, backlog, and dropping in the action-dependent queueing game. Both agents use independent one-hidden-layer tanh-softmax neural policies, so Fig. 4(a)–(b) directly evaluate neural minimax-learning convergence. Matched comparisons share the environment, policy architecture, initialization, seeds, entropy coefficient, and simultaneous start. The nominal benchmark compares GDA [15], the extragradient method (EGM) [10], PPM-3 as three inner fixed-point iterations approximating the proximal point method (PPM) [9], clipped minimax proximal policy optimization (Minimax PPO) [16], LCMPL-F with $\gamma_k = 0$, and LCMPL. Appendix B reports the complete protocol.

A. Neural Learning Dynamics and Policy Quality

The nominal study asks whether curvature selection improves neural learning and communication quality under moderate traffic. Fig. 4(a)–(b) show utility and exploitability over 60 updates; Fig. 4(c)–(d) report final goodput and backlog. Relative to LCMPL-F, LCMPL improves utility, exploitability, goodput, and backlog by 0.0072, 0.0213, 0.0039 packet/slot, and 0.0151 packet, respectively; every paired 95% interval is positive and 11 or 12 of 12 seeds improve. After disjoint-seed step-size selection, GDA, EGM, and PPM-3 are statistically indistinguishable from LCMPL, so the resolved ablation isolates the curvature benefit without relying on untuned baselines.

B. Load-Adaptive Communication Robustness

The second study tests whether the value of selective curvature grows as endogenous queueing stress tightens the service margin. Fig. 5 changes only the Bernoulli arrival rate and reports paired gains over LCMPL-F, with positive values preferred. The utility gain grows from an unresolved 0.0003 under light traffic to significant 0.0072 and 0.0121 gains under moderate and heavy traffic. At heavy load, goodput rises by 0.0069 packet/slot, while backlog and drop probability fall by 0.0090 packet and 0.0064; nonzero-curvature updates rise from 28% to 85% and 94% across the three loads. This load-dependent activation links the communication gains to selective geometric correction rather than to a permanently active extra direction.

VI. CONCLUSION

LCMPL separates field and curvature step sizes in queue-aware adversarial DSA and retains curvature only when the acceptance test validates merit and physical utility. The analysis bounds time-averaged stationarity and, under a local policy-gap condition, worst-case utility loss. The experiments show that this geometric stabilization improves packet delivery and limits backlog precisely where adversarial learning errors are most costly: near the queue’s service boundary.

APPENDIX A PROOFS

A. Proof of Proposition 1

With $\mathbf{A} = \nabla \mathbf{F}$, $\nabla \phi = \mathbf{A}^\top \mathbf{F}$ and $\mathbf{G} = \mathbf{A} \mathbf{F}$, hence $\langle \nabla \phi, \mathbf{G} \rangle = \mathbf{F}^\top \mathbf{A}^2 \mathbf{F}$. For $\mathbf{A} = \mathbf{S} + \mathbf{W}$, antisymmetry of $\mathbf{S}\mathbf{W} + \mathbf{W}\mathbf{S}$ removes the mixed term, leaving $\|\mathbf{S}\mathbf{F}\|^2 - \|\mathbf{W}\mathbf{F}\|^2$. $\square$

B. Proof of Proposition 2

For $i_k \in \arg \max_{i \in \{1,2\}} D_i(z_k)$, comparator feasibility, QP optimality, and Assumption 1(c) give $\mathbb{E}[\Delta V_k | \mathcal{F}_k] \leq -s \max\{D_F, D_G\} + \epsilon_k^{\text{mdl}} + \epsilon_k^{\text{acc}}$. Equations (11) and (12) complete the bound. $\square$

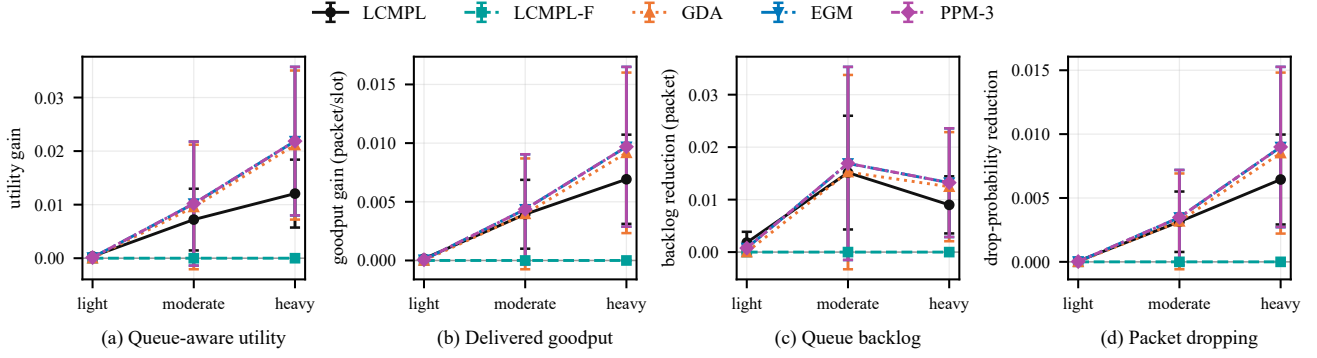


Fig. 5. Paired gains over LCMPL-F under traffic-load stress. Positive values favor the displayed method; error bars are 95% confidence intervals over 12 seeds. Light, moderate, and heavy loads use $\lambda_A = 0.20, 0.45$, and 0.70 .

C. Proof of Theorem 1

Taking expectations, summing Proposition 2, and telescoping yield $s\mu_{\text{geo}} \sum_{k < K} \mathbb{E} \|\mathbf{F}(\mathbf{z}_k)\|^2 \leq V(\mathbf{z}_0) - \mathbb{E} V(\mathbf{z}_K) + \sum_{k < K} \mathbb{E} \mathcal{E}_k(s)$; use $V(\mathbf{z}_K) \geq 0$ and divide by $s\mu_{\text{geo}} K$. $\square$

D. Proof of Corollary 1

For $v_\tau^* = \max_\pi \min_\nu J_\tau(\pi, \nu)$, $|J_\tau - J| \leq \tau \log M$ and $v_\tau^* \leq \max_\pi J_\tau(\pi, \nu_\psi)$ imply Eq. (14). Average it, bound Δ_τ , and apply Theorem 1. $\square$

APPENDIX B REPRODUCIBILITY DETAILS

The communication parameters are $M = L = 4$, $Q_{\max} = 4$, $\rho_h = 0.75$, $R_0 = N_0 = g_t^J = 1$, $I_t^{\text{pri}} = 0$, $(P_{ms}, P_{mJ}) = (10, 30)$, $c_{msw} = 0.05$, adjacent/nonadjacent leakage $(0.50, 0.08)$, $\delta = 0.95$, $\tau = 0.02$, and $(\alpha_q, \alpha_d) = (0.12, 0.60)$; the modes cyclically shift gains geometrically spaced from 1.9 to 0.30. Each tanh-softmax policy has 32 hidden units (452 parameters) and observes (g_t, q_t, a_{t-1}) ; constant I_t^{pri}, g_t^J are omitted and distinct g_t identifies h_t . Both studies use 60 updates, test seeds 800–811, $q_0 = 0$, and uniform h_0, a_{-1} ; nominal uses $\lambda_A = 0.45$ and six methods, whereas the load sweep uses $\lambda_A \in \{0.20, 0.45, 0.70\}$ without Minimax PPO. On validation seeds 100–103, GDA, EGM, and PPM-3 select 0.16 from $\{0.01, 0.04, 0.08, 0.16\}$; Minimax PPO uses step 10^{-3} , clip 0.2, and three epochs. LCMPL uses $[0, 0.16] \times [0, 0.12]$, eight backtracks, $(\lambda_\Delta, \lambda_U) = (0.35, 1.5)$, initial normalization, and the released tolerances.

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